

✓ **Praktikum Mandiri 09 | ML Senin Pagi**

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✓ **1. Import Library**

```
# Import Library
# Tahap ini mengimpor semua library Python yang diperlukan untuk analisis da
# pemodelan, dan evaluasi, sesuai dengan standar praktikum resmi Machine Lea

import pandas as pd                                # Membaca dan meman
import numpy as np                                  # Operasi numerik d
import matplotlib.pyplot as plt                    # Membuat visualisa
import seaborn as sns                              # Visualisasi data
from sklearn.model_selection import train_test_split # Membagi dataset m
from sklearn.preprocessing import StandardScaler    # Menstandarisasi f
from sklearn.naive_bayes import GaussianNB          # Model klasifikasi
from sklearn.metrics import accuracy_score, confusion_matrix, classification
from sklearn.model_selection import cross_val_score # Validasi silang
```

✓ **2. Loading Dataset**

```
# Menghubungkan colab dengan Google Drive
from google.colab import drive
drive.mount('/content/gdrive')
```

Drive already mounted at /content/gdrive; to attempt to forcibly remount, call

```
# Memanggil dataset lewat Gdrive
path = '/content/gdrive/MyDrive/Praktikum_ML/praktikum09/data/'
```

```
# Dataset yang digunakan adalah Breast Cancer Wisconsin (Diagnostic) dari UC
# Membaca file CSV menggunakan pandas
cancer_data = pd.read_csv(path + 'breast_cancer_wisconsin.csv')
```

```
# Menampilkan 5 baris pertama dataset
print("Lima baris pertama dataset:")
display(cancer_data.head())
```

Lima baris pertama dataset:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
0	842302	M	17.99	10.38	122.80	1001.0
1	842517	M	20.57	17.77	132.90	1326.0
2	84300903	M	19.69	21.25	130.00	1203.0
3	84348301	M	11.42	20.38	77.58	386.1
4	84358402	M	20.29	14.34	135.10	1297.0

5 rows × 33 columns

```
# Menampilkan 5 baris terakhir dataset
print("\nLima baris terakhir dataset:")
display(cancer_data.tail())
```

Lima baris terakhir dataset:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
564	926424	M	21.56	22.39	142.00	1479.0
565	926682	M	20.13	28.25	131.20	1261.0
566	926954	M	16.60	28.08	108.30	858.1
567	927241	M	20.60	29.33	140.10	1265.0
568	92751	B	7.76	24.54	47.92	181.0

5 rows × 33 columns

```
# Menampilkan dimensi dataset
print(f"\nDimensi dataset: {cancer_data.shape[0]} baris (entri) dan {cancer_
```

Dimensi dataset: 569 baris (entri) dan 33 kolom

✓ 3. Informasi Detail Dataset

```
print("Informasi detail dataset:")
cancer_data.info()
```

```
# - Total 569 entri
# - Terdapat kolom 'id' dan 'Unnamed: 32' yang tidak diperlukan
# - Kolom target: 'diagnosis' (M = malignant, B = benign)
# - Tidak ada missing value pada 31 kolom utama
```

```

Informasi detail dataset:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 33 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                    569 non-null    int64
1   diagnosis                            569 non-null    object
2   radius_mean                          569 non-null    float64
3   texture_mean                         569 non-null    float64
4   perimeter_mean                       569 non-null    float64
5   area_mean                           569 non-null    float64
6   smoothness_mean                      569 non-null    float64
7   compactness_mean                     569 non-null    float64
8   concavity_mean                       569 non-null    float64
9   concave points_mean                  569 non-null    float64
10  symmetry_mean                        569 non-null    float64
11  fractal_dimension_mean                569 non-null    float64
12  radius_se                             569 non-null    float64
13  texture_se                            569 non-null    float64
14  perimeter_se                          569 non-null    float64
15  area_se                              569 non-null    float64
16  smoothness_se                         569 non-null    float64
17  compactness_se                        569 non-null    float64
18  concavity_se                          569 non-null    float64
19  concave points_se                     569 non-null    float64
20  symmetry_se                           569 non-null    float64
21  fractal_dimension_se                  569 non-null    float64
22  radius_worst                          569 non-null    float64
23  texture_worst                         569 non-null    float64
24  perimeter_worst                       569 non-null    float64
25  area_worst                            569 non-null    float64
26  smoothness_worst                      569 non-null    float64
27  compactness_worst                     569 non-null    float64
28  concavity_worst                       569 non-null    float64
29  concave points_worst                  569 non-null    float64
30  symmetry_worst                        569 non-null    float64
31  fractal_dimension_worst                569 non-null    float64
32  Unnamed: 32                           0 non-null      float64
dtypes: float64(31), int64(1), object(1)
memory usage: 146.8+ KB

```

4. Missing Value & Pembersihan Data

```

# Cek jumlah missing value per kolom
print("Jumlah missing value per kolom:")
print(cancer_data.isnull().sum())

```

```

Jumlah missing value per kolom:
id                                0
diagnosis                         0
radius_mean                       0
texture_mean                       0
perimeter_mean                     0
area_mean                         0
smoothness_mean                    0
compactness_mean                    0

```

```
concavity_mean      0
concave points_mean  0
symmetry_mean       0
fractal_dimension_mean  0
radius_se           0
texture_se          0
perimeter_se        0
area_se             0
smoothness_se       0
compactness_se      0
concavity_se        0
concave points_se   0
symmetry_se         0
fractal_dimension_se  0
radius_worst        0
texture_worst       0
perimeter_worst     0
area_worst          0
smoothness_worst    0
compactness_worst   0
concavity_worst     0
concave points_worst  0
symmetry_worst      0
fractal_dimension_worst  0
Unnamed: 32         569
dtype: int64
```

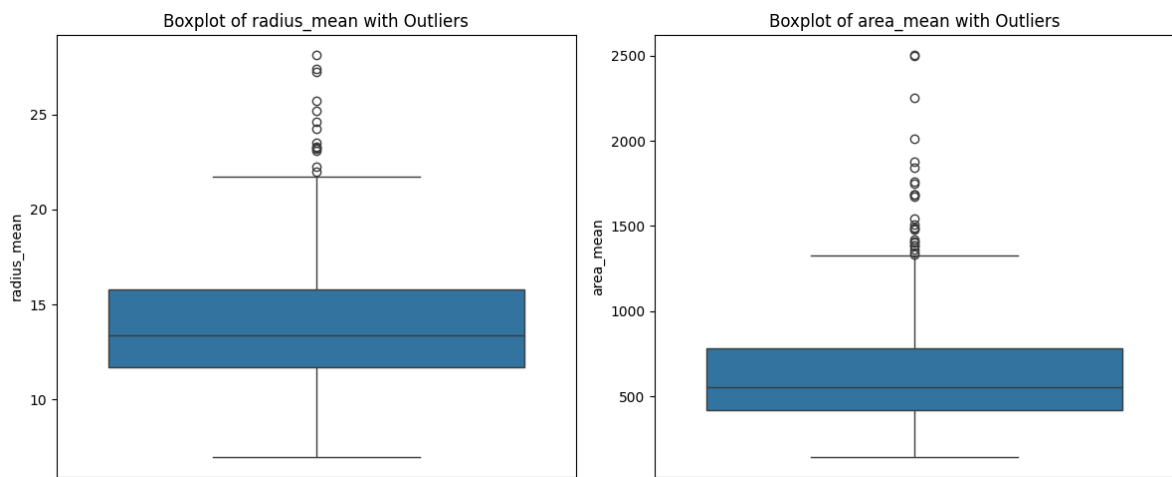
```
# Menghapus kolom yang tidak relevan untuk pemodelan
cancer_data = cancer_data.drop(columns=['Unnamed: 32', 'id'], errors='ignore')
```

```
# Cek dimensi setelah pembersihan
print(f"\nDimensi dataset setelah drop kolom tidak relevan: {cancer_data.sha
```

Dimensi dataset setelah drop kolom tidak relevan: (569, 31)

```
# Visualisasi outliers pada dua fitur utama
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
sns.boxplot(y=cancer_data['radius_mean'])
plt.title('Boxplot of radius_mean with Outliers')

plt.subplot(1,2,2)
sns.boxplot(y=cancer_data['area_mean'])
plt.title('Boxplot of area_mean with Outliers')
plt.tight_layout()
plt.show()
```



✓ 5. Data Analysis & Encoding Kategorikal

```
# Statistika deskriptif semua fitur numerik
print("Statistika Deskriptif Fitur Numerik:")
display(cancer_data.describe())
```

Statistika Deskriptif Fitur Numerik:

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
count	569.000000	569.000000	569.000000	569.000000	569.000000
mean	14.127292	19.289649	91.969033	654.889104	0.096360
std	3.524049	4.301036	24.298981	351.914129	0.014064
min	6.981000	9.710000	43.790000	143.500000	0.052630
25%	11.700000	16.170000	75.170000	420.300000	0.086370
50%	13.370000	18.840000	86.240000	551.100000	0.095870
75%	15.780000	21.800000	104.100000	782.700000	0.105300
max	28.110000	39.280000	188.500000	2501.000000	0.163400

8 rows × 30 columns

```
# Distribusi kelas target sebelum encoding
print("\nDistribusi kelas diagnosis sebelum encoding:")
print(cancer_data['diagnosis'].value_counts())
```

```
Distribusi kelas diagnosis sebelum encoding:
diagnosis
B      357
M      212
Name: count, dtype: int64
```

```
# Encoding kategorikal: Malignant (M) → 1, Benign (B) → 0
cancer_data['diagnosis'] = cancer_data['diagnosis'].map({'B': 0, 'M': 1})
```

```
# Distribusi setelah encoding
print("\nDistribusi kelas diagnosis setelah encoding:")
print(cancer_data['diagnosis'].value_counts())
```

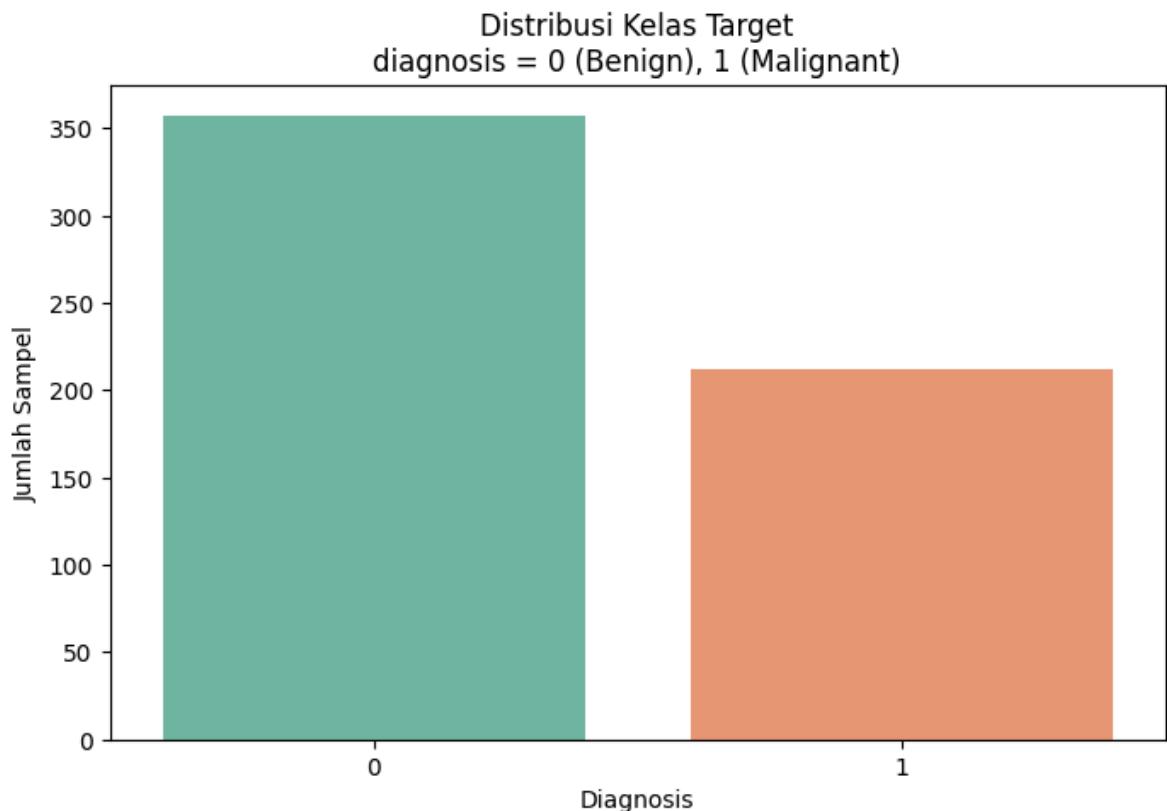
```
Distribusi kelas diagnosis setelah encoding:
diagnosis
0      357
1      212
Name: count, dtype: int64
```

```
# Visualisasi distribusi kelas target
plt.figure(figsize=(8,5))
sns.countplot(x='diagnosis', data=cancer_data, palette='Set2')
plt.title('Distribusi Kelas Target\ndiagnosis = 0 (Benign), 1 (Malignant)')
plt.xlabel('Diagnosis')
plt.ylabel('Jumlah Sampel')
plt.show()
```

```
/tmp/ipython-input-4003789370.py:3: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed i
```

```
sns.countplot(x='diagnosis', data=cancer_data, palette='Set2')
```



✓ 6. Visualisasi Hubungan Fitur dengan Target

```
# Menampilkan perbedaan distribusi fitur pada kedua kelas
```

```
fitur_visual = ['radius_mean', 'texture_mean', 'perimeter_mean',  
                'area_mean', 'concavity_mean', 'concave points_mean']
```

```
plt.figure(figsize=(15,10))
```

```
for i, col in enumerate(fitur_visual, 1):
```

```
    plt.subplot(2, 3, i)
```

```
    sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```

```
    plt.title(f'{col} vs Diagnosis')
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# Insight: Fitur-fitur ukuran dan bentuk tumor cenderung lebih tinggi pada k
```

```
/tmp/ipython-input-2782421541.py:9: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in

```
sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```

```
/tmp/ipython-input-2782421541.py:9: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in

```
sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```

```
/tmp/ipython-input-2782421541.py:9: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in

```
sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```

```
/tmp/ipython-input-2782421541.py:9: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in

```
sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```

```
/tmp/ipython-input-2782421541.py:9: FutureWarning:
```

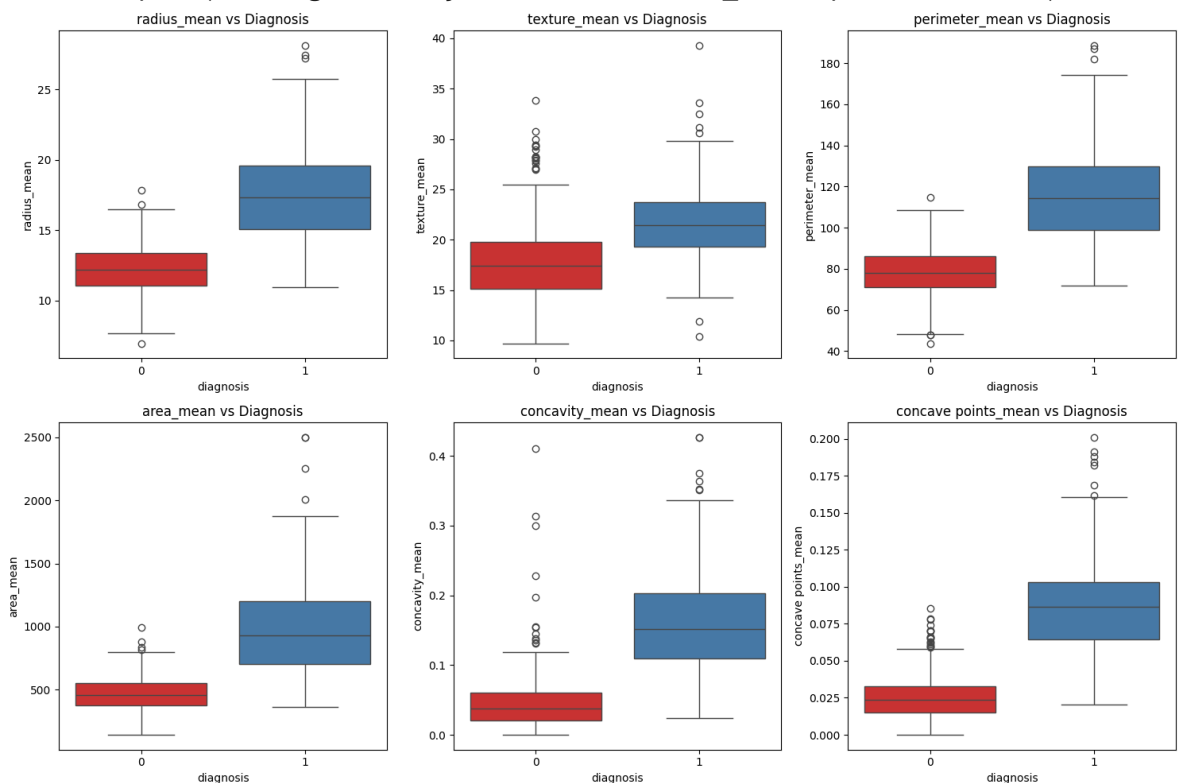
Passing `palette` without assigning `hue` is deprecated and will be removed in

```
sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```

```
/tmp/ipython-input-2782421541.py:9: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in

```
sns.boxplot(x='diagnosis', y=col, data=cancer_data, palette='Set1')
```



7. Separating, Splitting, and Scaling Data

```
# Memisahkan fitur dan target, membagi data, serta menstandarisasi fitur

# Pemisahan fitur (X) dan target (y)
X = cancer_data.drop(columns=['diagnosis']) # Semua kolom kecuali target
y = cancer_data['diagnosis']               # Kolom target

print(f"Ukuran matriks fitur X : {X.shape}")
print(f"Ukuran vektor target y : {y.shape}")
```

```
Ukuran matriks fitur X : (569, 30)
Ukuran vektor target y : (569,)
```

```
# Splitting data: 80% latih, 20% uji dengan stratify
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

print(f"\nJumlah data latih : {X_train.shape[0]} sampel")
print(f"Jumlah data uji : {X_test.shape[0]} sampel")
```

```
Jumlah data latih : 455 sampel
Jumlah data uji : 114 sampel
```

```
# Standarisasi fitur
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train) # Fit dan transform pada d
X_test_scaled = scaler.transform(X_test)      # Hanya transform pada dat
```

8. Naive Bayes Classification

```
# Pembuatan dan pelatihan model Gaussian Naïve Bayes
from sklearn.naive_bayes import GaussianNB

# Membuat instance model
nb_model = GaussianNB()

# Melatih model menggunakan data latih yang telah distandarisasi
nb_model.fit(X_train_scaled, y_train)
```

```
▼ GaussianNB ⓘ ?
GaussianNB()
```

9. Accuracy Score & Evaluation

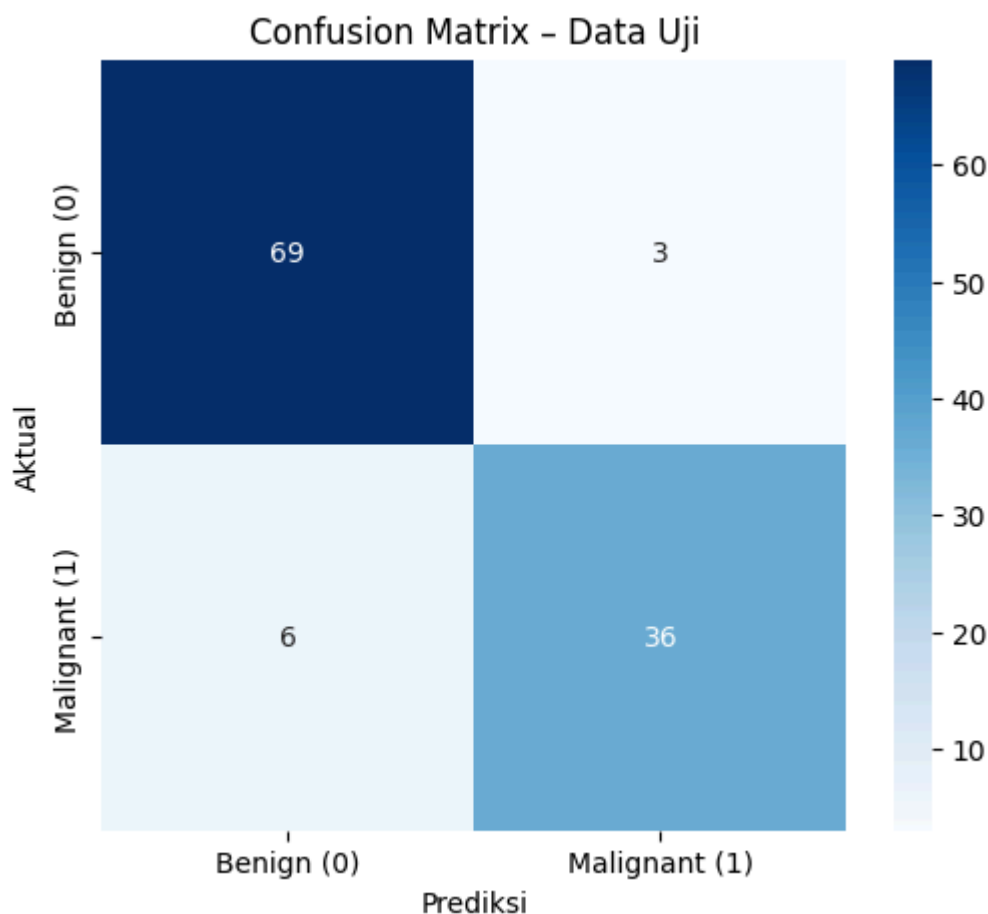
```
# Prediksi pada data latih dan data uji
y_train_pred = nb_model.predict(X_train_scaled)
y_test_pred = nb_model.predict(X_test_scaled)
```

```
# Menghitung akurasi
train_accuracy = accuracy_score(y_train, y_train_pred)
test_accuracy = accuracy_score(y_test, y_test_pred)
```

```
print(f"Akurasi pada data latih : {train_accuracy:.4f}")
print(f"Akurasi pada data uji   : {test_accuracy:.4f}")
```

```
Akurasi pada data latih : 0.9451
Akurasi pada data uji   : 0.9211
```

```
# Confusion Matrix
cm = confusion_matrix(y_test, y_test_pred)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Benign (0)', 'Malignant (1)'],
            yticklabels=['Benign (0)', 'Malignant (1)'])
plt.title('Confusion Matrix - Data Uji')
plt.xlabel('Prediksi')
plt.ylabel('Aktual')
plt.show()
```



```
# Classification Report
print("\nClassification Report (Data Uji):")
```

```
print(classification_report(y_test, y_test_pred,
                           target_names=['Benign (0)', 'Malignant (1)']))
```

```
Classification Report (Data Uji):
```

	precision	recall	f1-score	support
Benign (0)	0.92	0.96	0.94	72
Malignant (1)	0.92	0.86	0.89	42
accuracy			0.92	114
macro avg	0.92	0.91	0.91	114
weighted avg	0.92	0.92	0.92	114

✓ 10. Cross Validation (5-Fold)

```
# Validasi silang
cv_scores = cross_val_score(nb_model, X_train_scaled, y_train, cv=5, scoring

print("Hasil Cross-Validation (5-fold):")
for i, score in enumerate(cv_scores, 1):
    print(f"Fold {i}: {score:.4f}")
print(f"\nRata-rata akurasi : {cv_scores.mean():.4f}")
print(f"Standar deviasi    : {cv_scores.std():.4f}")
```

```
Hasil Cross-Validation (5-fold):
Fold 1: 0.9231
Fold 2: 0.9890
Fold 3: 0.9231
Fold 4: 0.9231
Fold 5: 0.9341
```

```
Rata-rata akurasi : 0.9385
Standar deviasi    : 0.0256
```

✓ Kesimpulan Tugas Mandiri Praktikum 09

1. Dataset Breast Cancer Wisconsin berhasil dimuat, dibersihkan, dan dianalisis secara eksploratori.
2. Tidak terdapat missing value signifikan. Kolom `id` dan `Unnamed: 32` dihapus karena tidak relevan.
3. Target `diagnosis` di-encode menjadi nilai biner (0 = Benign, 1 = Malignant).
4. Seluruh fitur numerik distandarisasi menggunakan `StandardScaler` karena GaussianNB sensitif terhadap skala data.
5. Model Gaussian Naïve Bayes dilatih dan dievaluasi dengan hasil sebagai berikut:
 - Akurasi data latih : 0.9451